Check for updates

Original research article

International Journal of Sports Science & Coaching

International Journal of Sports Science & Coaching
1–10
© The Author(s) 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/17479541251403877
journals.sagepub.com/home/spo
Sage

The perceptions and applications of cluster sets in strength and conditioning

Tsuyoshi Nagatani[1], Shayne Vial[1], Kristina L. Kendall[1], Paul Comfort[1,2], Jiahao Yang[1], and G. Gregory Haff[1,2,3]

Abstract

It is well documented that cluster sets, a modified set structure with additional rest between individual repetitions or groups of repetitions, can be used to modulate exercise-induced fatigue during resistance training. While practical guidelines exist, it remains unclear to what extent strength and conditioning (S&C) professionals utilise this programming tactic. The aim of this study was to examine S&C professionals' perceptions of cluster sets and their application in practice. A secondary aim was to assess the alignment between cluster set programming tactics used in the field and evidence-based recommendations from the scientific and coaching literature. Forty-six S&C professionals (age: 40.0 ± 13.6 yrs) completed an online survey comprised of three sections: (1) coaching demographic, (2) perceptions of cluster sets, and (3) program design. The survey included both closed- and open-ended questions. There was a general alignment between professional practice and current recommendations, particularly regarding the use of cluster sets during strength-power phases, manipulation of rest intervals, and prescription of higher training intensities. However, unlike recommended published practices, cluster sets were most commonly used with the bench press and variations of the squat, both of which require un-racking and re-racking the barbell during the cluster set, which may be physically demanding for athletes. Despite acknowledging logistical barriers such as supervision issues and time constraints, S&C professionals still view cluster sets as a practical programming tactic due to their minimal technological requirements and potential benefits. These findings can inform future applied research projects and educational content focused on cluster sets.

Keywords

Fatigue, inter-repetition rest, intra-set rest, professional practice, resistance training

Introduction

The physiological adaptations to resistance training can be altered by manipulating programming factors such as training intensity or volume, exercise selection, and order.[1] Another key programming factor that can be modified within resistance training programs is the structure of the training set.[2,3] Typically, there are two types of set structures that can be used when designing a resistance training program: a traditional set or a cluster set.[4,5] Traditional sets involve completing a series of repetitions in a continuous manner, with a pre-planned rest interval placed between sets (i.e., inter-set rest).[2,6] Alternatively, the cluster set involves an extended set structure where short rest periods are introduced between individual repetitions (i.e., inter-repetition rest) or groups of repetitions (i.e., intra-set rest) within the set.[7] It is well documented that cluster sets can be used to maintain barbell velocity during resistance training exercises, compared to traditional sets, where there are typically significant declines in velocity and greater mechanical strain.[8–12] These changes to mechanical stress are often accompanied by changes in acute physiological responses, with the traditional set eliciting elevations in both blood lactate and growth hormone when compared to cluster sets.[13]

Reviewer: Hunter Bennett (University of South Australia, Australia)

[1]Strength and Power Research Group, School of Medical and Health Science, Edith Cowan University, Joondalup, Western Australia
[2]Directorate of Psychology and Sport, University of Salford, Salford, UK
[3]Exercise Medicine Research Institute, Edith Cowan University, Joondalup, Western Australia

Corresponding author:
Tsuyoshi Nagatani, Edith Cowan University, School of Medical and Health Science, 270 Joondalup Drive, Joondalup, Western Australia 6027, Australia.
Email: t.nagatani@ecu.edu.au

Based upon the noted differences in acute mechanical and physiological responses between traditional and cluster sets, it has been suggested that the cluster set be used in training phases where the primary goal of training is the improvement of the athlete's rate of force development.[14,15] On the other hand, traditional sets may be better suited for training phases where improvements in local muscle endurance or muscle hypertrophy is the primary training goal.[14,15] Additionally, several alternative cluster set configurations (See Figure 1) have been proposed as advanced programming tactics when targeting an increased training variability.[4,14] These recommendations are based upon several training studies that have examined the difference between chronic training adaptations associated with the performance of traditional and cluster sets.[16–18] While practical guidelines on the use of cluster sets are presented in the scientific and coaching literature,[14,15] the prevalence with which these programming tactics are utilised by strength and conditioning (S&C) professionals remains unknown. The lack of clarity about the use of cluster sets highlights a potential gap between research-based recommendation and real-world practice.

While acute performance and fatigue responses, as well as chronic training adaptations to resistance training performed using cluster sets have been extensively investigated,[2,19] much of the current research pertaining to cluster sets focuses on non-athletic populations and employs training protocols that may not be representative of those commonly used in a high-performance environment.[9,20,21] To develop more ecologically valid and relevant study designs that can better guide professional practice, it is essential to appreciate the practical knowledge of S&C professionals and address real-world performance challenges that will inform daily practice and decision-making among end users and stakeholders.[22] Therefore, the aim of this study was to investigate S&C professionals' perceptions about the use of cluster sets and to determine how they implement cluster sets within their professional practice. This study was also used to examine the extent to which cluster set programming tactics used within professional practice align with the recommendations provided in the scientific and coaching literature.

Methods

Participants

Forty-six S&C professionals (39 males and 7 females, means ± SD; age: 40.0 ± 13.6 years, working experience within the S&C discipline: 12.3 ± 10.0 yrs) participated in this study. Participants were recruited using email networks and various social media platforms such as Instagram, Facebook, X and LinkedIn. To broaden participant recruitment, a chain sampling method was used, in which respondents were encouraged to pass on details of the survey to other S&C professionals.[23] Eligible participants were required to be actively serving or had past experience as a S&C coach, having worked with athletes at the professional, amateur club, collegiate, national, international, and/or Olympic levels. To maximise the sample size and capture a broad range of coaching practices, no restrictions were applied to inclusion criteria based on coaching experience or accreditation level. Prior to completing the questionnaire, electronic informed consent was collected, in accordance with the Edith Cowan University Human Research Ethics Committee (Project Number 2024-05253).

Procedures

The survey was conducted using an online survey platform (Qualtrics, Provo, UT). Participants were asked to answer closed-ended questions in the form of single and multiple-choice formats. Open-ended questions provided the participants with the opportunity to elaborate on their answers to specific questions regarding the perceptions of implementing cluster sets. The questionnaire consisted of three main sections including: (1) coaching demographics, (2) perceptions of cluster sets, and (3) program design. The 'coach's demographic' section was designed to explore the backgrounds of the participants. In the 'perceptions of cluster sets' section, the focus of questions was placed on the participant's knowledge regarding the application of cluster sets during resistance training. The last part of this questionnaire focused on exploring how participants program cluster sets within resistance training programs. The electronic questionnaire took approximately 10–15 min to complete. The whole survey is included as supplementary material.

Statistical analyses

Response data were exported to an Excel spreadsheet from the web based survey software (Qualtrics, Provo, UT) and analysed with the R statistical programming language (Version 4.2.2).[24] Analysis procedures included frequency counts and percentage calculations for all closed-ended questions. Responses to open-ended questions were assessed using a six-stage thematic analysis[25] that included (i) familiarisation with the data, (ii) generating initial codes, (iii) searching for themes, (iv) reviewing themes, (v) defining and naming themes, (vi) producing the report. This approach allows clear and distict overarching themes representing the main ideas or patterns to be generated from the raw data for each open-ended question.[25] In some cases, responses to open-ended questions provided enough detail to identify more than one overarching theme.

Results

Characteristics of respondents

There were 46 respondents who met the inclusion criteria (39 males and 7 females, means ± SD; age: 40.0 ± 13.6 years,

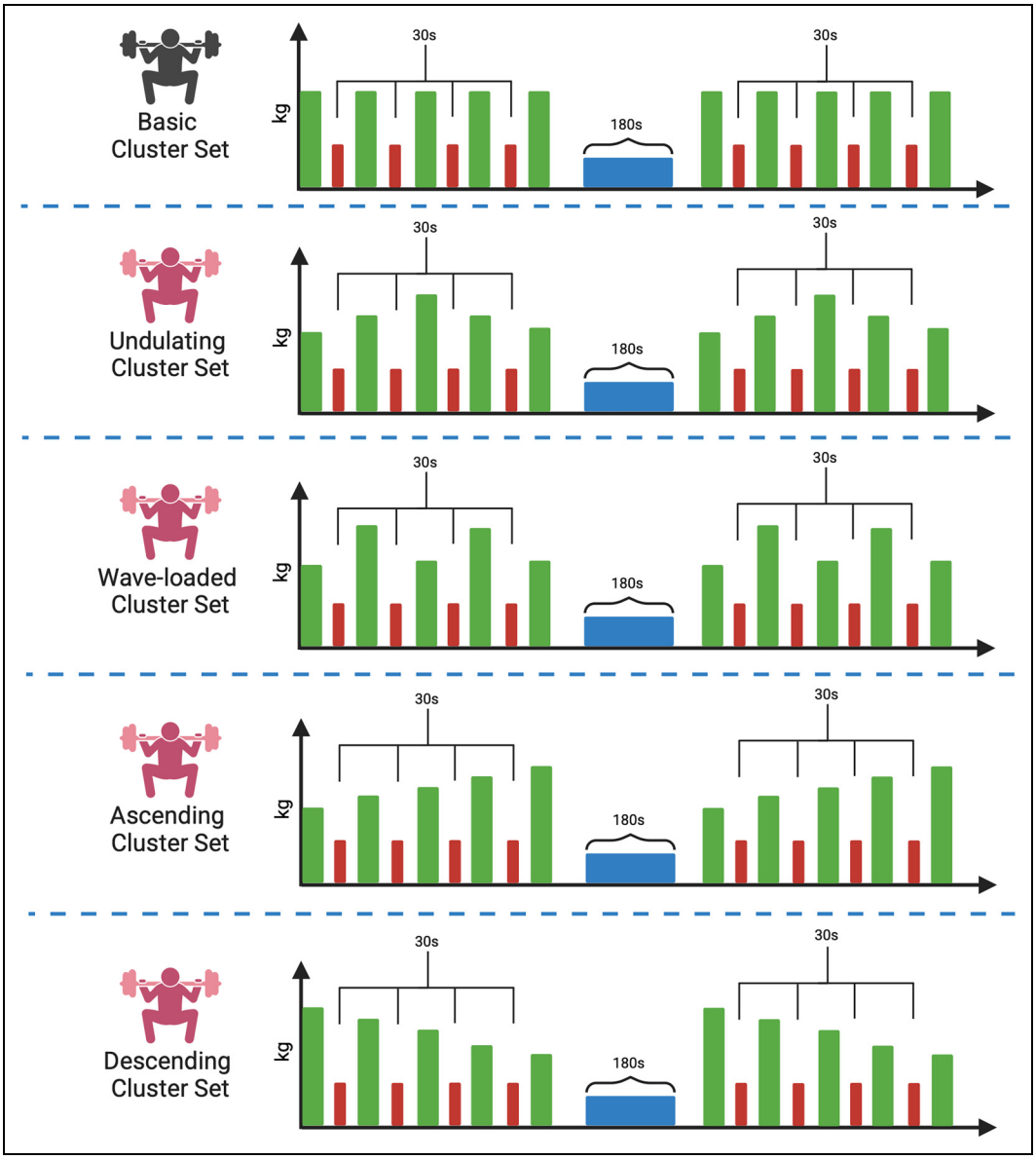

Figure 1. Examples of basic and alternative cluster set structures when using 2 sets of 5 repetitions. Created in BioRender. NAGATANI, T. (2025) https://BioRender.com/kshvr7t. Adapted from Haff[49].

working experience within the S&C discipline: 12.3 ± 10.0 yrs). Their current jobs included: S&C coach (n = 22, 48%), Head S&C coach (n = 13, 28%), High Performance Manager (n = 5, 11%), Academic Staff (n = 5, 11%), Retired (n = 1, 2%). These roles were based within various professional contexts including: sports institutes (n = 6, 13%), professional/semiprofessional sports club (n = 16, 35%), higher education or collegiate system (n = 19, 41%), amateur sports club (n = 22, 48%) or private sector (n = 1, 2%). The S&C-specific qualifications held by participants included accreditations from profesisonal bodies such as National Strength and Conditioning Association (n = 27, 59%), Australian Strength and Conditioning Association (n = 15, 33%), United Kingdom Strength and Conditioning Association (n = 3, 7%), Collegiate Strength and Conditioning Coaches Association (n = 1, 2%). Specific academic S&C qualifications held by participants included BSc (n = 6, 13%), MSc (n = 25, 54%), PhD (n = 9, 20%) or none (n = 6, 13%).

Applications of cluster sets

Thirty-seven respondents (80%) reported that they used cluster sets as a programming tactic. Among those who used cluster sets, sixteen respondents reported prescribing one or more of the alternative cluster set structures (Figure 1). Respondents who had not used cluster sets (n = 9/46) reported the following reasons for not using these

set structures: supervision issues due to large athlete training groups (n = 5, 56%), unconvinced about the effectiveness of this training method (n = 4, 44%), increased total training time due to added rest (n = 3, 33%), lack of necessity for use (n = 2, 22%), limited knowledge about this training method (n = 1, 11%), lack of scientific evidence supporting its use (n = 1, 11%) or never having heard of this training method (n = 1, 11%).

Respondents who reported using cluster sets (n = 37) stated that they used them in the hypertrophy/strength-endurance (n = 13, 35%), basic strength (n = 18, 49%), and strength-power (n = 31, 84%) phases of a periodised training plan. Methods that S&C professionals reported using include the percentage of 1RM (n = 23, 62%), a repetition maximum (n = 7, 19%), load-velocity profiles (n = 7, 19%) or repetitions in reserve (n = 1, 3%) to determine training load when prescribing cluster sets. Based on the open-ended question, most respondents reported using a percentage of 1RM and increasing training intensity in a similar manner (Table 1). Respondents reported that they generally programmed multi-joint upper- and lower-body exercises when prescribing cluster sets (Figure 2). A summary of the key training variables manipulated when prescribing cluster sets with each training phase is presented in Figure 3. The sources of information that respondents currently use or have used to inform the prescription of cluster sets are summarised in Table 2.

Perceptions of cluster sets

While approximately half of the respondents who reported using cluster sets (n = 19/37) had no concerns regarding their implementation, the remaining respondents (n = 18) expressed several concerns, which are summarised in Table 3. A large portion of respondents (n = 29/37) rated themselves as "*confident*" to "*very confident*" in their knowledge and ability to program cluster sets. Similarly, a large portion of respondents (n = 27/37) reported understanding the underpinning science associated with programming cluster sets "*well*" to "*very well*". Key themes were developed based on the qualitative responses to open-ended questions about what S&C practitioners would like to know to improve their confidence in programming cluster sets. The number of responses to each theme and exempler responses are provided in Table 4.

Table 1. Responses to how strength and conditioning coaches increase the training load when you program cluster sets.

Exemplar Responses
"The range of intensity increase would be between 5–10%"
"Increase load by approximately 10–15%"
"While I do increase 5 to 10%, these increases take place over a 4 week meso cycle"
"I will have the athletes lifting 5–15% more for cluster sets compared to the regular programming. Stronger athletes can handle the extra intensity"
"Usually linear increase of 2–5% depending on the experience and aptitude of the lifter"
"Usually I prescribe 7.5–10% 1RM higher in a cluster set than I would typically prescribe in a straight set of the same number of repetitions,"
"Generally, 5–10%, depending on the training phase, athlete's technical abilities, and competitive schedule"

Discussion

The aim of this study was to gain insights into the perceptions of cluster sets and their implementation by S&C professionals. We also aimed to examine the extent to which cluster set programming strategies used in professional practice align with the recommendations provided in the scientific and coaching literature. In the present investigation, a large portion of S&C professionals (i.e., n = 37/46) reported incorporating cluster sets into their athletes' resistance training programs. However, several S&C professionals highlighted practical challenges, particularly the difficulty of supervising large groups of athletes performing cluster sets in high-performance environments. These logistical issues are especially relevant in team sport settings, where overseeing multiple athletes simultaneously can be difficult. Despite these challenges, cluster sets remain more accessible than other advanced programming methods, such as velocity-based training,[26] because they do not require specialised technology or equipment. As such, cluster sets represent a relatively straightforward programming tactic for S&C professionals, provided that the logistical barriers to implementation can be managed.

When integrating cluster sets into a resistance training program, it is important to establish the training goal(s) of the athlete within the context of a periodised training plan, ensuring that targeted training outcomes align with these established goals.[14] In the present study, most S&C professionals reported prescribing cluster sets during the strength-power phase (84%), while fewer applied them in hypertrophy/strength-endurance phases (35%). These practices align with the current cluster set programming recommendations presented in the scientific and coaching literature.[14,19] For example, hypertrophy/strength-endurance training typically emphasise high training volumes at low to moderate training relative loads to promote muscle growth and increase work capacity, thereby preparing athletes for subsequent training phases.[27] In contrast, strength-power training generally use lower training volumes using a mixed method approach, with exercises and loads targeting multiple points along the load-velocity curve.[28] In support of these programming recommendations, a meta-analysis by Jukic et al.[19] highlighted that traditional sets had more favourable training effects on local muscle endurance improvement compared to cluster sets, whereas cluster sets induced greater velocity and power outputs at various submaximal loads compared to

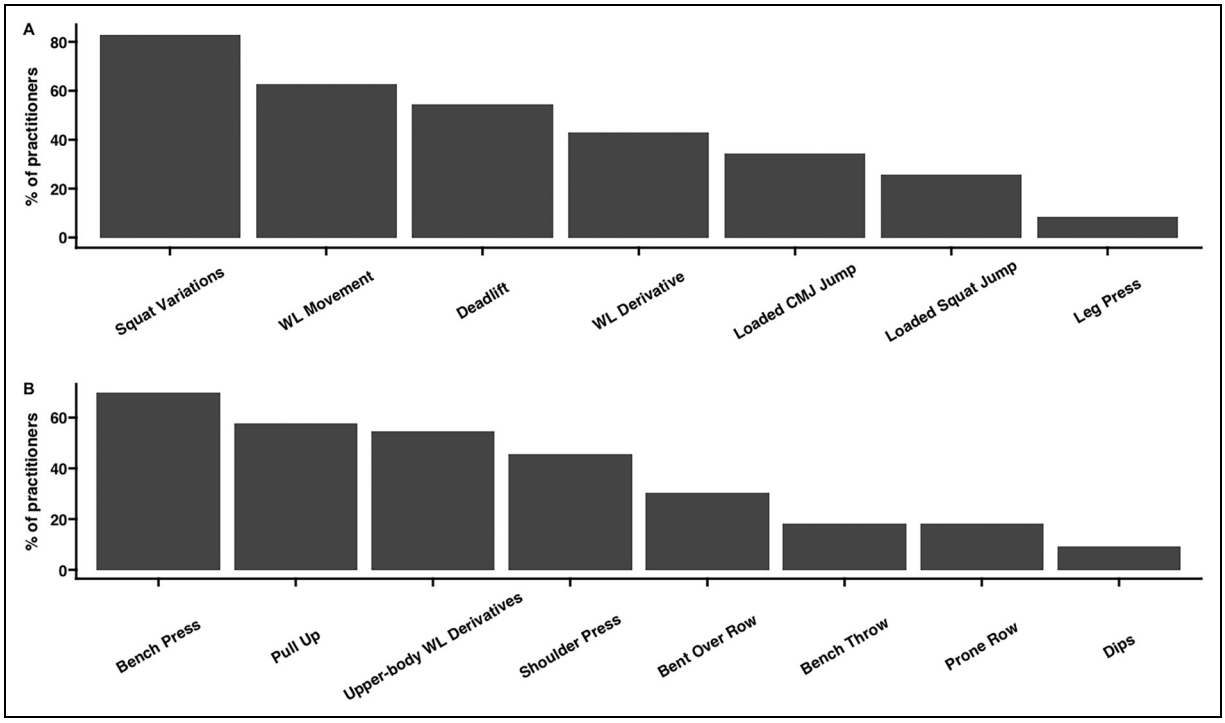

Figure 2. A summary of exercise used when programming cluster sets.
Note: (A) Lower-body resistance exercises and (B) upper-body resistance exercises. CMJ = Countermovement jump; WL = Weightlifting.

traditional sets. Therefore, it appears that S&C professionals are applying cluster sets in a manner that is consistent with the current evidence-based recommendations. This is unsurprising since academic journals (70%) were noted as the primary source of information by the respondents to this survey. This finding highlights the important role that current scientific and coaching literature play in shaping the practices of S&C professionals.

When designing resistance training programs, applying variation to training variables is crucial, as these modifications influence the physiological adaptations to training.[1] Training variables relevant to cluster sets include exercise selection, training intensity, intra-set and inter-repetition rest interval, and the number of sets and repetitions.[14] These variables should be adjusted in accordance with the training goals defined by the periodised training plan.[14] This study has highlighted that S&C professionals modify several training variables depending on the specific training phase (Figure 3). For example, S&C practitioners tended to use a high repetition range (9–12 repetitions) with relatively short intra-set rest (6–15 s) and short inter-repetition rest (5–15 s) in the hypertrophy/strength-endurance phase. Conversely, training programs in the strength-power phase tended to consist of a low repetition range ($\leq$ 5 repetitions) with relatively long intra-set (16–30 s) combined with either short (11–15 s) or long ($\geq$ 26 s) inter-repetition rest. These regimes seemed to be performed 1–2 times a week for three or four weeks (i.e., one meso-cycle). It is important to note, however, that the extent to which the regimes in the current study are evidence-based and the exact training history of the athletes who performed these programs remain unknown. While this information would be valuable, it was beyond the scope of the questionnaire. As such, if readers consider using this information to inform training prescription, caution should be exercised.

In the present study, bench press and squat variants (e.g., goblet squat, back squat, front squat, etc.) were the most common exercises used when programming cluster sets (Figure 2). This is not surprising given that these movements are among the most frequently investigated in the current body of scientific literature pertaining to the effects of cluster sets, and academic journals were reported to be a primary source of information about cluster sets.[2,19] However, the benefits from using cluster sets may not be fully realised when implemented with the bench press and variants of the squat as these exercises require un-racking and re-racking the barbell between each repetition or groups of repetitions contained within a cluster set.[15] This practice can be physically demanding for the athlete, requiring a considerable effort to get into the right starting position before performing the movement,[29] unless a specialised squat rack (i.e., monolift) is available, which allows the athlete to un-rack the barbell and initiate the movement on the spot.[14] Given these challenges, cluster sets may have more

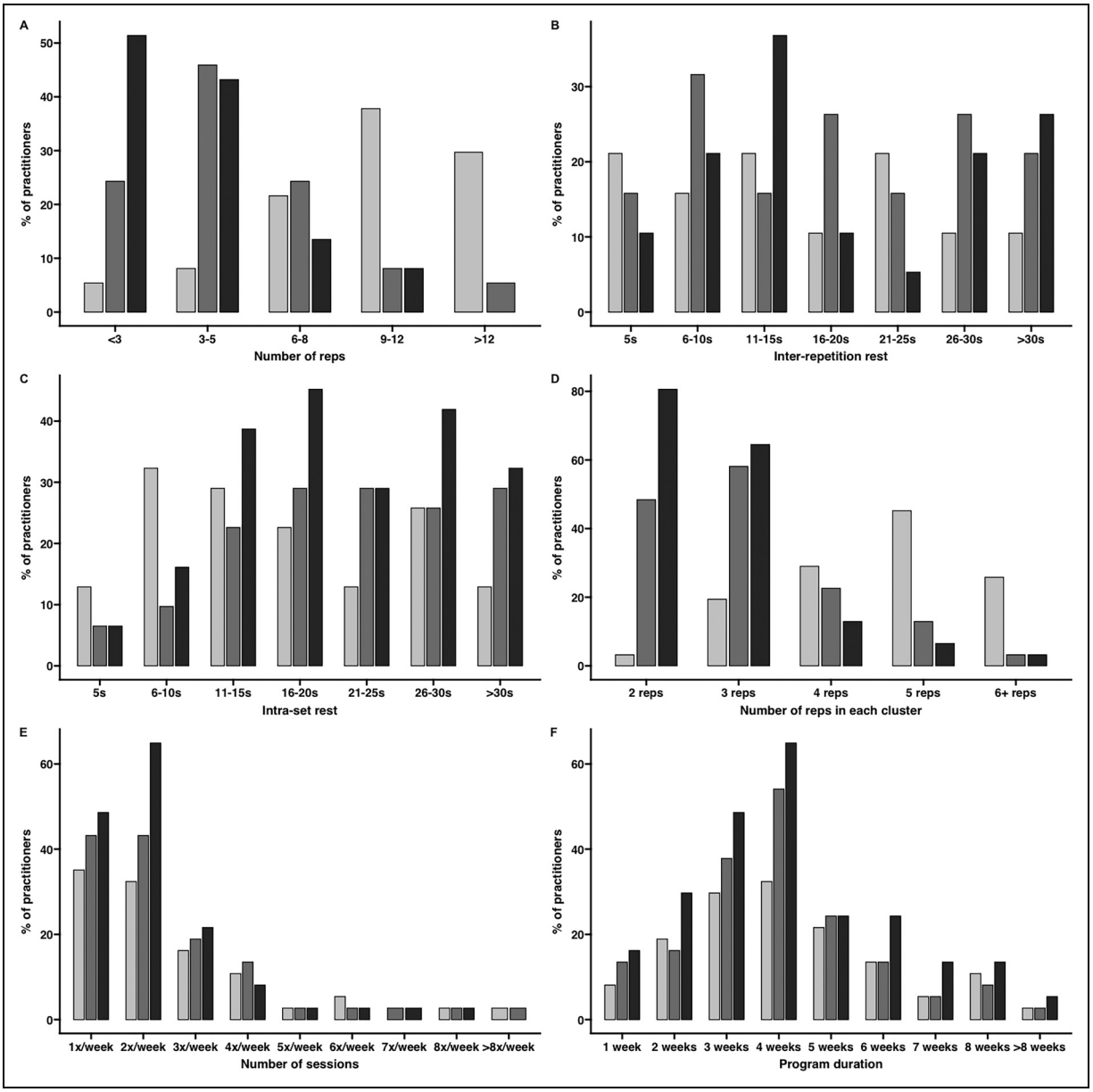

Figure 3. A summary of the key training variables manipulated when prescribing cluster sets with each training phase.
Note: Programs designed for the hypertrophy/strength-endurance, basic strength and strength-power phases are represented by light grey, grey and dark grey bars, respectively. (A) Number of repetitions in a training set, (B) duration of inter-repetition rest, (C) duration of intra-set rest, (D) number of repetitions in the clusters when prescribing cluster sets with intra-set rest intervals, (E) number of sessions per week and (F) duration of exercise program.

practicality with exercises that do not involve re-racking the barbell such as weightlifting movements and their derivatives.[15] If, however, readers choose to implement cluster sets with the bench press and squat variations, it is recommended to extend the intra-set or inter-repetition rest intervals in order to accommodate the additional effort and time required for re-racking and set-up.

Monitoring and adjusting training intensity based on an athlete's current strength levels is essential to develop muscular strength. This biomotor capacity has been reported to underpin many key athletic performance tasks (e.g., jumping, sprinting, and change of direction).[30–33] There are several methods that can be used to regularly assess and adjust training intensity for strength development.[34] Among these, the two most frequently used by S&C professionals are prescribing training intensity as a percentage of an athlete's 1RM or using repetition maximum zones, where the heaviest load is lifted for a given repetition range.[34] Findings from the current investigation confirmed that these two methods are also the most commonly employed when prescribing

Table 2. Responses to what resources strength and conditioning coaches have used to inform their programming of cluster sets during resistance training.

Rank	Type of Resource	Percentage of Respondents
1	Academic journals	26/37 (70%)
2	S&C experts/practitioners	24/37 (65%)
3	Personal experience	23/37 (62%)
4	S&C colleagues	22/37 (60%)
5	Conferences	17/37 (46%)
6	books	15/37 (41%)
7	workshops	11/37 (30%)
8	Internet websites	11/37 (30%)
9	Trade journals/magazines	6/37 (16%)
10	Social media	6/37 (16%)
11	Podcasts	6/37 (16%)
12	Sport coaches	5/37 (14%)
13	Powerlifting coaches	4/37 (11%)

Note: S&C = Strength and conditioning

Table 3. Responses to what concerns strength and conditioning coaches have regarding implementing cluster sets.

Rank	Concern	Percentage of Respondents
1	No concern	19/37 (51%)
2	Extended training duration from added rest	10/37 (27%)
3	supervision issues due to large athlete training groups	9/37 (24%)
4	disapproval from coaches	4/37 (11%)
5	inappropriate athlete population	2/37 (5%)
6	lacking knowledge about this training method	1/37 (3%)
7	lack of scientific evidence supporting its use	1/37 (3%)

training intensity for cluster sets. Additionally, S&C professionals reported prescribing higher training intensity for cluster sets compared to traditional sets. Specifically, practitioners increase training intensity by 2–15% of 1RM when implementing cluster sets (Table 1). This practice is supported by several researchers who have reported that cluster sets allow for the use of higher training intensities and/or volumes during resistance exercises compared to traditional sets.[12,35] Given that high-intensity resistance training has been shown to be more effective for developing maximal strength compared to low- to moderate-intensity resistance training,[36,37] it is recommended that higher training intensities be used when programming cluster sets in order to maximise strength development.

A main concern expressed by S&C practitioners about the application of cluster sets is the extended training duration associated with the added rest used to implement the cluster set. For example, González-Hernández et al.[10] reported that cluster sets with the inclusion of 30 s inter-repetition rest during 3 sets of 10 back squats resulted in better maintenance of movement velocity but significantly increased session time compared to traditional sets (25.0 vs 11.5 min). As time is often limited in real-world settings, this programming method may not be practical for some S&C professionals. To accommodate the limited time available for strength training, several authors have proposed an alternative approach that is referred to as a rest-redistribution set.[38,39] Rest-redistribution sets involve redistributing rest periods within the set without changing overall training time.[6] Several researchers have reported that rest-redistribution sets are a more effective method than traditional sets for maintaining mechanical performance,[40–42] lessening metabolic stress[13,43] and perceptual exertion.[44,45] Although cluster sets and rest-redistribution sets are often considered interchangeable in the literature,[13,40–43] it is important to distinguish between them, as they may result in different training outcomes. In a meta-analysis by Jukic et al.,[2] it was reported that rest-redistribution sets are less effective than cluster sets at maintaining movement velocity during resistance training and this is likely due to the abbreviated rest periods in rest-redistribution sets, which may not provide sufficient time for adequate recovery of adenosine triphosphate and phosphocreatine. This incomplete recovery could, in turn, lead to different acute metabolic responses and potentially distinct long-term training adaptations. This contention is supported by a meta-analytic study reporting that alternative set structures, including rest-redistribution sets, can lead to greater improvements in vertical jump performance, velocity and power at submaximal loads when compared with traditional, although comparative results have been observed for muscular hypertrophy and maximal strength gains.[19] Despite growing interest in rest-redistribution sets as a time-efficient alternative,[21,39] the extent to which S&C professionals implement this method in applied settings remands unknown. Therefore, future research should explore the prevalence and practical applications of this programming strategy in professional practice.

While the results of this investigation provide valuable insights into S&C professionals' perceptions about the use of cluster sets and how they implement them in their professional practice, several limitations should be acknowledged. To maximise the sample size and capture a broad range of coaching practices, no restrictions were applied to inclusion criteria based on coaching experience. However, it is plausible that more experienced S&C professionals (e.g., ≥ 10 years of coaching experience) may hold different perceptions regarding the use of cluster sets and implement them differently compared with their less experienced counterparts (e.g., ≤ 3 years of coaching experience). Additionally, the data collected via questionnaire may not be as comprehensive as that obtained through more in-depth qualitative methods, such as face-to-face interviews. For example, a semi-structured interview, where researchers use a pre-planned guide to direct the interaction and rely predominantly on open-ended questions, allows the participant a certain degree of flexibility to express their opinions,

Table 4. Responses to what strength and conditioning coaches would like to know to improve their confidence in programming cluster sets.

Rank	Theme	Exemplar Responses	Percentage of Respondents
1	Programming strategies	"Programming intensity and appropriate rest intervals within different intensities" and "Weekly undulating of cluster volume and intensity"	15/46 (33%)
2	Nothing	"Nothing" and "I feel quite comfortable using cluster sets depending on the block of training"	15/46 (33%)
3	Periodisation strategies	"How to implement cluster sets in different ways throughout the periodisation process", "When to use them (cluster sets)", "When the period is best to use this method during the season", and "At what point in a training program do clusters best fit?"	9/46 (20%)
4	Targeted adaptations and outcomes	"Practical guidelines to using cluster sets for different desired adaptations based on evidence in population with advanced training age (strength, power, hypertrophy etc.)"	7/46 (15%)
5	Population-specific application	"Programming (cluster sets) with youth athletes who have low training ages", "Application for bigger groups of athletes", and "Programming cluster sets for different sports"	5/46 (11%)
6	Mechanisms and scientific evidence	"What the mechanisms behind the positive adaptations are" and "Is there scientific backing to support the use of cluster sets?"	5/46 (11%)
7	Educational tools and knowledge sharing	"I know how and why I apply them but would appreciate how others do so" and "Infographics and media may help build more education and buy in for athletes"	2/46 (4%)

Note: PAPE = Post-activation performance enhancement

ideas, and experiences in greater depth.[46] This approach has been used by several researchers to enhance the quality and depth of data collected, which allows for a comprehensive understanding of the meanings that S&C professionals attach to their practices, specifically the rationale behind implementing cluster sets.[47,48] Therefore, future study aiming to explore specific aspects of applied practice should consider using semi-structured intervews to provide more nuanced insights into a given topic area.

Conclusion

The current study provided valuable insights into how S&C professionals perceive and apply cluster sets in an athlete's resistance training program. These findings indicate a general alignment between professional practice and current evidence-based recommendations, particularly regarding the use of cluster sets during strength-power phases, manipulation of rest intervals, and prescription of higher training intensities. Despite acknowledging logistical barriers such as supervision issues and time constraints, S&C professionals still view cluster sets as a practical programming tactic due to their minimal technological requirements and potential benefits. Additionally, although rest-redistribution sets may offer a time-efficient alternative to cluster sets, further research is needed to elucidate their application and effectiveness in professional settings. Overall, this study contributes to bridging the gap between research and practice by proving insight into the application of cluster sets in real-world settings. These findings can inform future applied research projects and educational content that can impact the practices of S&C professionals operating within high-performance environments.

Acknowledgements

The authors would like to thank all participants who partook in this study for their time.

ORCID iDs

Tsuyoshi Nagatani https://orcid.org/0000-0003-2187-157X
Shayne Vial https://orcid.org/0000-0002-9235-8979
Kristina L. Kendall https://orcid.org/0000-0001-6872-7335
Paul Comfort https://orcid.org/0000-0002-1131-8626
Jiahao Yang https://orcid.org/0000-0001-5153-9098
G. Gregory Haff https://orcid.org/0000-0002-0676-7750

Ethical considerations

The ethical approval was granted by the Institution's Human Research Ethics Committee (Project 2023-04209).

Consent to participate

Prior to completing the questionnaire, electronic informed consent was obstained by each participant, in accordance with the Human Research Ethics Committee.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supplemental material

Supplemental material for this article is available online.

References

1. Kraemer WJ, Ratamess NA and French DN. Resistance training for health and performance. *Curr Sports Med Rep* 2002; 1: 165–171.
2. Jukic I, Ramos AG, Helms ER, et al. Acute effects of cluster and rest redistribution set structures on mechanical, metabolic, and perceptual fatigue during and after resistance training: a systematic review and meta-analysis. *Sports Med* 2020; 50: 2209–2236.
3. Haff GG, Whitley A, McCoy LB, et al. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res* 2003; 17: 95–103.
4. Haff GG, Hobbs RT, Haff EE, et al. Cluster training: a novel method for introducing training program variation. *Strength Cond J* 2008; 30: 67–76.
5. Haff GG, Burgess S and Stone M. Cluster training: theoretical and practical applications for the strength and conditioning professional. *Prof Strength Cond* 2008: 12–17.
6. Tufano JJ, Brown LE and Haff GG. Theoretical and practical aspects of different cluster set structures: a systematic review. *J Strength Cond Res* 2017; 31: 848–867.
7. Verkhoshansky Y and Siff MC. *Supertraining*. Rome, Italy: Verkhoshansky SSTM, 2009.
8. Tufano JJ, Conlon JA, Nimphius S, et al. Maintenance of velocity and power with cluster sets during high-volume back squats. *Int J Sports Physiol Perform* 2016; 11: 885–892. DOI: org/10.1123/ijspp.2015-0602.
9. García-Ramos A, González-Hernández JM, Baños-Pelegrín E, et al. Mechanical and metabolic responses to traditional and cluster set configurations in the bench press exercise. *J Strength Cond Res* 2020; 34: 663–670.
10. González-Hernández JM, García-Ramos A, Castaño-Zambudio A, et al. Mechanical, metabolic, and perceptual acute responses to different set configurations in full squat. *J Strength Cond Res* 2020; 34: 1581–1590.
11. Wagle JP, Cunanan AJ, Carroll KM, et al. Accentuated eccentric loading and cluster set configurations in the back squat: a kinetic and kinematic analysis. *J Strength Cond Res* 2018; 35: 420–427.
12. Tufano JJ, Conlon JA, Nimphius S, et al. Cluster sets: permitting greater mechanical stress without decreasing relative velocity. *Int J Sports Physiol Perform* 2017; 12: 463–469.
13. Oliver JM, Kreutzer A, Jenke S, et al. Acute response to cluster sets in trained and untrained men. *Eur J Appl Physiol* 2015; 115: 2383–2393.
14. Nagatani T, Haff GG, Guppy SN, et al. Practical application of traditional and cluster set configurations within a resistance training program. *Strength Cond J* 2022; 44: 87–101.
15. Nagatani T, Kendall KL, Guppy SN, et al. Using cluster set configurations within a resistance training programme. *Prof Strength Cond* 2022: 7–17.
16. Goto K, Ishii N, Kizuka T, et al. The impact of metabolic stress on hormonal responses and muscular adaptations. *Med Sci Sports Exerc* 2005; 37: 955–963.
17. Arazi H, Khoshnoud A, Asadi A, et al. The effect of resistance training set configuration on strength and muscular performance adaptations in male powerlifters. *Sci Rep* 2021; 11: 1–10.
18. Zaras N, Stasinaki A-N, Spiliopoulou P, et al. Effect of inter-repetition rest vs. Traditional strength training on lower body strength, rate of force development, and muscle architecture. *Appl Sci* 2021; 11: 45–56.
19. Jukic I, Van Hooren B, Ramos AG, et al. The effects of set structure manipulation on chronic adaptations to resistance training: a systematic review and meta-analysis. *Sports Med* 2021; 51: 1061–1086.
20. Cuevas-Aburto J, Jukic I, Chirosa-Ríos LJ, et al. Effect of traditional, cluster, and rest redistribution set configurations on neuromuscular and perceptual responses during strength-oriented resistance training. *J Strength Cond Res* 2020; 36: 1490–1497.
21. Iacono AD, Watson K and Jukic I. The autoregulation rest-redistribution training method mitigates sex differences in neuromuscular and perceived fatigue during resistance training. *Int J Sports Physiol Perform* 2024; 19: 685–695.
22. Harden M, Bruce C, Wolf A, et al. Exploring the practical knowledge of eccentric resistance training in high-performance strength and conditioning practitioners. *Int J Sports Sci Coach* 2020; 15: 41–52.
23. Suri H. Purposeful sampling in qualitative research synthesis. *Qual Res J* 2011; 11: 63–75.
24. R Core Team. *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing, 2024.
25. Nowell LS, Norris JM, White DE, et al. Thematic analysis: Striving to meet the trustworthiness criteria. *Int J Qual Methods* 2017; 16.
26. Weakley J, Mann B, Banyard H, et al. Velocity-based training: from theory to application. *Strength Cond J* 2021; 43: 31–49.
27. Stone MH, Hornsby WG, Haff GG, et al. Periodization and block periodization in sports: emphasis on strength-power training - A provocative and challenging narrative. *J Strength Cond Res* 2021; 35: 2351–2371.
28. Haff GG and Nimphius S. Training principles for power. *Strength Cond J* 2012; 34: 2–12.
29. Davies TB, Tran DL, Hogan CM, et al. Chronic effects of altering resistance training set configurations using cluster

sets: a systematic review and meta-analysis. *Sports Med* 2021; 51: 707–736.
30. Suchomel TJ, Nimphius S and Stone MH. The importance of muscular strength in athletic performance. *Sports Med* 2016; 46: 1419–1449.
31. Weyand PG, Sternlight DB, Bellizzi MJ, et al. Faster top running speeds are achieved with greater ground forces not more rapid leg movements. *J Appl Physiol* 2000; 89: 1991–1999.
32. Hori N, Newton RU, Andrews WA, et al. Does performance of hang power clean differentiate performance of jumping, sprinting, and changing of direction? *J Strength Cond Res* 2008; 22: 412–418.
33. Nimphius S, McGuigan MR and Newton RU. Changes in muscle architecture and performance during a competitive season in female softball players. *J Strength Cond Res* 2012; 26: 2655–2666.
34. Suchomel TJ, Nimphius S, Bellon CR, et al. Training for muscular strength: methods for monitoring and adjusting training intensity. *Sports Med* 2021; 51: 2051–2066.
35. Iglesias E, Boullosa DA, Dopico X, et al. Analysis of factors that influence the maximum number of repetitions in two upper-body resistance exercises: curl biceps and bench press. *J Strength Cond Res* 2010; 24: 1566–1572.
36. Schoenfeld BJ, Grgic J, Ogborn D, et al. Strength and hypertrophy adaptations between low-vs. High-load resistance training: a systematic review and meta-analysis. *J Strength Cond Res* 2017; 31: 3508–3523.
37. Lasevicius T, Ugrinowitsch C, Schoenfeld BJ, et al. Effects of different intensities of resistance training with equated volume load on muscle strength and hypertrophy. *Eur J Sport Sci* 2018; 18: 772–780.
38. Jukic I and Tufano JJ. Shorter but more frequent rest periods: no effect on velocity and power compared to traditional sets not performed to failure. *J Hum Kinet* 2019; 66: 257–268.
39. Meechan D, McMahon JJ, Suchomel TJ, et al. The effect of rest redistribution on kinetic and kinematic variables during the countermovement shrug. *J Strength Cond Res* 2023; 37: 1358–1366.
40. Oliver JM, Kreutzer A, Jenke SC, et al. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res* 2016; 30: 235–243.
41. Oliver J, Jenke S, Mata J, et al. Acute effect of cluster and traditional set configurations on myokines associated with hypertrophy. *Int J Sports Med* 2016; 37: 1019–1024.
42. Hansen KT, Cronin JB and Newton MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform* 2011; 6: 455–468.
43. Girman JC, Jones MT, Matthews TD, et al. Acute effects of a cluster-set protocol on hormonal, metabolic and performance measures in resistance-trained males. *Eur J Sport Sci* 2014; 14: 151–159.
44. Ho IM, Luk JT, Ngo JK, et al. Effects of different intraset rest durations on lifting performance and self-perceived exertion during bench press exercise. *J Strength Cond Res* 2019; 35: 2114–2120.
45. Boffey D, Clark NW and Fukuda DH. Efficacy of rest redistribution during squats: considerations for strength and sex. *J Strength Cond Res* 2021; 35: 586–595.
46. Sparkes AC and Smith B. *Data collection. Qualitative research methods in sport, exercise, and health: from process to product*. New York: Routledge, 2014, pp.83–114.
47. Holmberg PM, Russell S, O'Brien KA, et al. Exploring strength and conditioning practitioners' perceptions about using priming exercise as a pre-competition strategy to improve performance. *Int J Sports Sci Coach* 2024; 19: 1598–1611.
48. Thompson SW, Olusoga P, Rogerson D, et al. Is it a slow day or a go day?": the perceptions and applications of velocity-based training within elite strength and conditioning. *Int J Sports Sci Coach* 2023; 18: 1217–1228.
49. Haff GG. *Strength, power, and hypertrophy training. Scientific foundation and practical application of periodization*. Champaign, United States: Human Kinetics, 2025, pp.335-382.